UniMind ■ - Milestone Execution Report

Step 1: Scaffold Project Structure & Environment

Project: AI-Powered University Knowledge Assistant | **Team:** Group G21 (Utkarsh, Suhani, Sanchayeeta)
Status: COMPLETED & VERIFIED (3/3 Tests Passed) | **Date:** November 1, 2026

1. Deliverables & Constraint Adherence

Deliverable / Constraint	Status	Verification Details
Project Directory Structure	COMPLETED	Clean modular architecture created (backend, frontend, data, vector_store, scripts, tests, docs).
Backend & Frontend Placeholders	COMPLETED	Created backend/app structure and frontend/README.md placeholder.
requirements.txt	COMPLETED	Pinned core API, data parsing, vector search, LLM, and test tools.
.env.example	COMPLETED	Template created with ports, CORS origins, data paths, and model configs.
FastAPI Application Skeleton	COMPLETED	Modular app with Settings, CORS, root endpoint, /health, /api/v1/health.
.gitignore Verification	PRESERVED	Already present; ignores .env, .venv, cache; left unmodified.
Technical Documentation	COMPLETED	Created ARCHITECTURE.md, SETUP_GUIDE.md, API_SPECIFICATION.md.
No RAG / Gemini / React Yet	STRICTLY ADHERED	No premature logic, fake documents, or unnecessary dependencies.

2. Created Files Register

File / Directory	Type	Purpose / Description
.env.example	Config	Configuration template for ports, CORS origins, and future model keys.
requirements.txt	Dependencies	Specifies FastAPI, Uvicorn, Pydantic, PyMuPDF, FAISS, Sentence Transformers, Gemini.
backend/app/main.py	Python Entrypoint	FastAPI application instance, CORS middleware, lifespan context, and root endpoints.
backend/app/core/config.py	Configuration	12-factor configuration management loading .env using Pydantic BaseSettings.
backend/app/models/schemas.py	Data Models	Pydantic DTO models (HealthResponse, RootResponse).
backend/app/routes/health.py	Router	Health check endpoint handler (GET /health).
backend/app/routes/api.py	Router Aggregator	API v1 router grouping versioned endpoints under /api/v1 prefix.
backend/app/services/__init__.py	Package	Placeholder package for future business logic and orchestrations.
backend/app/rag/__init__.py	Package	Placeholder package for ingestion, chunking, embeddings, and grounding logic.
backend/app/utils/__init__.py	Package	Placeholder package for common utilities and helpers.
backend/README.md	Documentation	Backend structure guide, running instructions, and swagger URLs.
frontend/README.md	Documentation	Frontend architecture placeholder for React + Tailwind student UI.
data/raw/.gitkeep	Marker	Tracks directory for original university policy PDFs and notices.
data/processed/.gitkeep	Marker	Tracks directory for normalized and chunked text documents.
data/evaluation/.gitkeep	Marker	Tracks directory for golden Q&A; evaluation datasets.
vector_store/.gitkeep	Marker	Tracks directory for persisted local FAISS vector index files.
scripts/.gitkeep	Marker	Tracks directory for indexing and evaluation automation scripts.
tests/test_health.py	Test Suite	Automated unit tests covering root endpoint and health check endpoints.
documentation/ARCHITECTURE.md	Documentation	In-depth architecture specification, diagrams, layer roles, and abstention rules.

documentation/SETUP_GUIDE.md	Documentation	Step-by-step developer setup, virtualenv creation, and test execution instructions.
documentation/API_SPECIFICATION.md	Documentation	Formal REST endpoint specifications for current and future phases.

3. Automated Testing & Verification Results

The FastAPI skeleton was verified by running unit tests against the endpoints using the built-in test client:

```
Command: python -m unittest tests/test_health.py
Ran: 3 unit tests in 0.019s
Result: PASS (OK)
• test_root_endpoint: HTTP 200, status='online', service name, documentation URLs validated.
• test_root_health_check: HTTP 200, status='healthy', environment & version validated.
• test_api_v1_health_check: HTTP 200, versioned router check (/api/v1/health) validated.
```

4. Git Status & Next Milestone Ready

Git status was verified to ensure zero unintended alterations to existing repository files (.gitignore, LICENSE, README.md). All new files remain cleanly staged as untracked additions awaiting user approval.

Next Milestone Available: Step 2 – Document Ingestion, PDF Processing & Chunking Pipeline.